05-29 第2篇

阅读理解（说明文） How Do Birds Find Their Way Home Across Thousands of Miles?

① Every autumn, swans leave the cold lakes of northern China and fly south, sometimes to Poyang Lake or even further. Every spring, they return to almost exactly the same lake where they were born. Some small birds fly more than 10,000 kilometres between their summer home and winter home, crossing mountains, deserts and seas. They carry no map and no phone. **How do they know the way?** Scientists have been working on this mystery for many years, and now we know at least four pieces of the answer.

② First, birds read the **sun** during the day. They do not only see it as a source of light; they also remember the **time** of day. If the sun is on their left at nine in the morning, they know they are flying south. A built-in clock in their small brains tells them how the sun's position changes during the hours. In this way, the sun becomes a slow, moving compass.

③ Second, at night, birds look at the **stars**. Scientists have tested young songbirds in a special round room called a "planetarium". When the fake stars were shown in their normal positions, the birds flew towards the correct direction. When the star pattern was turned around, the birds quickly changed their direction too. This showed that many night-flying birds, like geese and some small robins, use the stars as a sky map.

④ Third, birds can feel the **Earth's magnetic field** — something that humans cannot feel at all. Our planet is like a very big, very soft magnet. Tiny magnetic particles in the beaks or eyes of some birds, such as the European robin, help them understand where north and south are. Even on a cloudy night without stars, these birds can still fly in the right direction. This ability is often called a "magnetic sense".

⑤ Fourth, birds remember **landmarks**. A young swan, during its first migration, flies with older members of its family. Every year, they learn new mountain ranges, rivers and coastlines. Scientists have put tiny tracking devices on some birds and found that many of them follow almost the same curved path every year, turning left at the same river mouth and right above the same mountain pass. Just like a taxi driver in a big city, experienced birds know the small corners of the sky by heart.

⑥ Taken together, the sun, the stars, the magnetic field and familiar landmarks make birds **natural navigators**. It is a system that does not need batteries, satellites or Wi-Fi. Next time a group of birds flies overhead in a tidy V-shape, look up, and remember:

each of them carries a living compass inside, made of light, time, magnetism and memory.

根据短文内容，选择最佳答案。

1. What is the passage mainly about?

A. Why birds build nests.
B. How birds find their way across very long distances.
C. How to make a paper compass.
D. Why humans like to watch birds.

2. How do birds use the sun, according to paragraph ②?

A. They believe the sun is a kind of fruit.
B. They use only the sun's colour.
C. They use the sun's position together with an inside clock, so the sun becomes a slow, moving compass.
D. They never use the sun.

3. Which experiment showed that birds use the stars?

A. Tests with young songbirds in a round room called a "planetarium".
B. Tests with birds in the sea.
C. Tests that put birds inside cardboard boxes.
D. Tests with birds flying in complete darkness outside.

4. What is the "magnetic sense" of birds?

A. Their ability to sing.
B. Their ability to feel the Earth's magnetic field and know where north and south are.
C. Their ability to count.
D. Their ability to read books.

5. How do birds learn landmarks?

A. They read maps given by scientists.
B. A young swan flies with older family members during its first migration and learns mountains, rivers and coastlines.
C. They fly alone from day one and guess the way.
D. They follow paper signs placed by farmers.

答案与解析

1.
【答案】B
【题型】主旨大意题
【定位】全文围绕鸟类长途导航
【解析】文章讲鸟类如何在数千公里距离内找路。A/C/D 均非主题。故选 B。

2.
【答案】C
【题型】细节理解题
【定位】第②段 "A built-in clock in their small brains tells them how the sun's position changes..."
【解析】太阳+内置时钟=移动罗盘。A/B/D 均错。故选 C。

3.
【答案】A
【题型】细节理解题
【定位】第③段 "Scientists have tested young songbirds in a special round room called a 'planetarium'."
【解析】天文馆实验。B/C/D 均无据。故选 A。

4.
【答案】B
【题型】词义猜测题 / 细节理解题
【定位】第④段 "feel the Earth's magnetic field... understand where north and south are."
【解析】感知地磁。A/C/D 均无据。故选 B。

5.
【答案】B
【题型】细节理解题
【定位】第⑤段 "A young swan... flies with older members of its family."
【解析】随长辈学地标。A/C/D 均错。故选 B。

词汇注释（超纲词）

- **compass** [n.] 罗盘；指南针
- **magnetic field** [n.] 磁场
- **migration** [n.] 迁徙
- **landmark** [n.] 地标
- **navigator** [n.] 导航者